



10962 - Peering Into the Heart of the Angel Nebula

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam Observations of M2CC				
	1	M2CC - NIRCam	NIRCam Imaging	(1) M2CC
MIRI Observations of M2CC				
	2	M2CC - MIRI	MIRI Imaging	(1) M2CC
	3	M2CC - MIRI (central arcmin)	MIRI Imaging	(1) M2CC

ABSTRACT

Star formation laws govern our understanding of the production of stars from a volume of gas, with the well-known Kennicutt-Schmidt relation standing the testament of both time and scale across multiple types of galaxies at different epochs. In our own Galaxy, a landmark study recently discovered a global, linear relationship between the star formation rate (SFR) and the gas surface density divided by the free-fall time of that gas, implying a near-constant free-fall efficiency across Milky Way clouds of all shapes and sizes. This efficiency, however, drops quickly at the highest column densities, precisely where star formation should be at its most efficient. The cause? There are two options: observational biases leading to inaccurate counts of the present protostars; or increasing feedback making it more difficult to form stars at the highest densities. Here, we propose to observe the heart of the Angel Nebula to conclusively determine which of these cases it is. The Angel Nebula has the most compact high column density region within which to efficiently map and answer this question. Only JWST has the infrared sensitivities, resolutions, and wavelength coverage necessary to identify the most embedded protostar populations and their feedback in Galactic young stellar clusters. Together, these will determine how a molecular cloud's most recent star-formation activity directly relates to its current physical characteristics. These data will provide wealth beyond even this important science question, with practical use in substellar object formation research, and outflow research. As a final bonus, these data will produce rich and informative figures for public outreach.

OBSERVING DESCRIPTION

The proposed project includes three imaging mosaics of the Mon R2 central cluster (M2CC), an embedded cluster of young stellar objects (YSOs) at 860 pc distance. The M2CC is compact and rich and its high column density regions extend over just 6' x 6', making it an excellent target for JWST imaging. The goal is to obtain a census of the most highly embedded YSOs in the M2CC, to determine if the local star forming efficiency in the cloud truly declines with gas column density there or not, and what influence feedback has on the compact environment. Hence, multi-band imaging with both NIRCam and MIRI are required to identify YSOs via their spectral energy distributions.

The first mosaic is composed of 2 x 2 pointings of NIRCam, which will use a FULLBOX dither type with both modules in a 4TIGHT primary pattern and two subdithers to cover the M2CC. The cluster will be observed in two medium-band NIRCam SW/LW filter pairs, F162M + F480M and F182M + F460M, and a narrow-band filter pair F187N + F405N, with 4 x 1 pointings.

The second mosaic is composed of 3 x 5 pointings of MIRI using its full array. This mosaic is nearly identical in size to the NIRCam one, enabling nearly-complete coverage at up to ten wavelengths. These observations will use the CYCLING dither type and the LARGE pattern. The M2CC will be observed in four wide-band MIRI filters, F1130W, F1280W, F1500W and F2100W.

The third mosaic is composed of 2 x 1 pointings of MIRI using its SUB256 subarray over the central two arcminutes of the observations. This

JWST Proposal 10962 (Created: Monday, April 27, 2026, 3:00:30PM Eastern Standard Time) - Overview

mosaic is necessary given the bright nebulosity in the central arcminute of the M2CC around the source Mon R2 IRS 1 at mid-infrared wavelengths. The shorter integrations enabled by using SUB256 will prevent saturation over this specific location. These observations will also use the CYCLING dither type and the LARGE pattern. The central region of the M2CC will also be observed in same four wide-band filters as the second mosaic, F1130W, F1280W, F1500W and F2100W.

There are no special restrictions.

Proposal 10962 - Targets - Peering Into the Heart of the Angel Nebula

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	M2CC	RA: 06 07 46.3000 (91.9429167d) Dec: -06 22 54.00 (-6.38167d) Equinox: J2000	Epoch of Position: 2000	
Comments: Category=Stellar Cluster Description=[Protoclusters, Young star clusters]					

Proposal 10962 - Observation 1 - Peering Into the Heart of the Angel Nebula

Observation	Proposal 10962, Observation 1: M2CC - NIRCcam									Mon Apr 27 20:00:30 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCcam Imaging										
Diagnostics	(M2CC - NIRCcam (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.									
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	M2CC	RA: 06 07 46.3000 (91.9429167d) Dec: -06 22 54.00 (-6.38167d) Equinox: J2000			Epoch of Position: 2000				
	Comments: Category=Stellar Cluster Description=[Protoclusters, Young star clusters]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order
	3	1	8.0		8.0		0.0		0.0	DEFAULT
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		4TIGHT		SMALL-GRID-DITHER				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F162M+F150W2	F480M	RAPID	2	2	16	8	429.471	273080
	2	F182M	F410M	RAPID	2	2	16	8	429.471	273080
	3	F187N	F405N+F444W	MEDIUMDEEP2	4	1	8	8	4037.026	273080

Proposal 10962 - Observation 1 - Peering Into the Heart of the Angel Nebula

Special Requirements	Sequence Visits within 53.0 Days
	Aperture PA Range 40.92542306 to 90.92542306 Degrees (V3 41.0 to 91.0)
	Aperture PA Range 103.92542306 to 104.92542306 Degrees (V3 104.0 to 105.0)
	Aperture PA Range 116.92542306 to 123.92542306 Degrees (V3 117.0 to 124.0)
	Aperture PA Range 129.92542306 to 133.92542306 Degrees (V3 130.0 to 134.0)
	Aperture PA Range 137.92542306 to 170.92542306 Degrees (V3 138.0 to 171.0)
	Aperture PA Range 180.92542306 to 207.92542306 Degrees (V3 181.0 to 208.0)
	Aperture PA Range 217.92542306 to 224.92542306 Degrees (V3 218.0 to 225.0)
	Aperture PA Range 234.92542306 to 257.92542306 Degrees (V3 235.0 to 258.0)
	Aperture PA Range 262.92542306 to 265.92542306 Degrees (V3 263.0 to 266.0)
	Aperture PA Range 282.92542306 to 305.92542306 Degrees (V3 283.0 to 306.0)
	Aperture PA Range 323.92542306 to 352.92542306 Degrees (V3 324.0 to 353.0)
	Aperture PA Range 359.92542306 to 6.92542306 Degrees (V3 0.0 to 7.0)
	Visits Same PA

Proposal 10962 - Observation 2 - Peering Into the Heart of the Angel Nebula

Observation	Proposal 10962, Observation 2: M2CC - MIRI										Mon Apr 27 20:00:30 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:13) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:14) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:15) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:16) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:17) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:18) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:19) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:20) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
(1)		M2CC	RA: 06 07 46.3000 (91.9429167d) Dec: -06 22 54.00 (-6.38167d) Equinox: J2000			Epoch of Position: 2000					
Comments: Category=Stellar Cluster Description=[Protoclusters, Young star clusters]											
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	4	5	5.0		5.0		0.0		0.0		HILBERT_CURVE
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						LARGE	

Proposal 10962 - Observation 2 - Peering Into the Heart of the Angel Nebula

Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1130W	FASTR1	5	3	1	Dither 1	4	12	188.703	273080
	2	F1800W	FASTR1	5	3	1	Dither 1	4	12	188.703	273080
	3	F1500W	FASTR1	5	3	1	Dither 1	4	12	188.703	273080
	4	F2100W	FASTR1	5	3	1	Dither 1	4	12	188.703	273080
Special Requirements	Group Visits within 53.0 Days Visits Same PA										

Proposal 10962 - Observation 3 - Peering Into the Heart of the Angel Nebula

Observation	Proposal 10962, Observation 3: M2CC - MIRI (central arcmin)										Mon Apr 27 20:00:30 GMT 2026
	Diagnostic Status: Warning Observing Template: MIRI Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(1)	M2CC	RA: 06 07 46.3000 (91.9429167d) Dec: -06 22 54.00 (-6.38167d) Equinox: J2000				Epoch of Position: 2000				
	Comments: Category=Stellar Cluster Description=[Protoclusters, Young star clusters]										
Template	Subarray										
	SUB256										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	5	4	10.0		10.0		0.0		0.0		DEFAULT
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1130W	FASTR1	5	25	1	Dither 1	4	100	178.514	273080
	2	F1800W	FASTR1	5	25	1	Dither 1	4	100	178.514	273080
	3	F1500W	FASTR1	5	25	1	Dither 1	4	100	178.514	273080
	4	F2100W	FASTR1	5	25	1	Dither 1	4	100	178.514	273080
Special Requirements	Group Visits within 53.0 Days Visits Same PA										